Hidden architecture and the thermodynamic cost of an exact stochastic signal

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Abstract

When only part of a system can be watched, one model-based way to bound its dissipation is to fit the smallest hidden Markov model that explains the data and minimise entropy production over that model. We show by an exact example that this procedure can overstate the least possible dissipation, even when the observed signal is known perfectly. We prove that increasing the number of hidden states can strictly reduce the least entropy production compatible with an entire observed stochastic process. For a binary alternating-renewal family, we determine the global minimum over all three-state continuous-time Markov realisations: the homogeneous driven ring is the unique minimiser up to relabelling, with cost $(1-q)\log(1/q)$. For every $0 < q < 1$, we construct a finite four-state exact realisation with strictly smaller cost. At $q = 1/20$, a simple generator with rational rates reproduces the same complete stationary observation law at cost $2.773076657835\ldots$, below the three-state minimum $2.845945659876\ldots$. Thus minimal state count does not certify the thermodynamic minimum, even with exact observations. We also solve the unrestricted optimisation constrained by the entry intensity r, mean complete dwell m, and initial dwell density $f(0)$ of one observed block. For $0 < mf(0) < 1$ the exact infimum is $4r\sqrt{1-mf(0)}\,\operatorname{artanh}\sqrt{1-mf(0)}$; it is approached by three-state models with diverging internal rates but is attained by no finite model. For $mf(0) \geq 1$ a reversible model attains zero cost. The resulting bound applies without renewal independence and places the exact-law benchmark's all-architecture infimum between $1.897297106584\ldots$ and $2.746985402969\ldots$, the certified cost of an exact rational five-state model; the lower end is about 108 times the floor previously certified for this benchmark in an earlier Evidence Press candidate. Its exact value and finite-dimensional attainment remain unresolved. Numerical exploration, which is not part of the proofs, indicates that the four-state optimum is about 2.765996 and that a fifth state lowers the cost further.

All theorems have explicit proofs and reproducible exact-arithmetic certificates.

1 Introduction

The question in plain terms. Many single-molecule experiments record only a coarse signal: an ion channel is open or shut, a motor is bound or unbound, a nanopore is blocked or clear. The microscopic mechanism behind the signal is a hidden Markov network whose size is unknown. Its entropy production, the thermodynamic cost of keeping it out of equilibrium, cannot be read off the signal directly. One model-based route is to fit hidden models that reproduce the signal and report the least dissipation among them. That least value is a valid lower bound on the true cost only if the true mechanism belongs to the fitted class. This paper asks whether choosing the *smallest* exactly fitting class is safe. It is not: we exhibit a signal, known exactly, whose cheapest four-state mechanism dissipates strictly less than every three-state mechanism, even though three states already reproduce the signal perfectly.

Inferring dissipation from a partially observed system requires a choice of admissible microscopic models. If the hidden network is known, one can minimise entropy production over the rates compatible with the observations. If its size is unknown, an additional optimisation over hidden architecture is necessary. An exact fit with the fewest states solves a representation problem. Whether it also solves the thermodynamic problem is a separate mathematical question. Allowing more states can never raise the minimum, because a state can be split into copies with proportional fluxes; the content of the question is whether the minimum can strictly *fall*.

Existing inference methods make this distinction consequential. Ehrich's optimisation over hidden Markov models uses all observed statistics but assumes the number of hidden states is known, and explicitly asks how adding hidden states or cycles affects the bound [1]. Nitzan, Ghosal and Bisker combine transition statistics with waiting-time moments, and investigate simpler assumed topologies numerically [2]. Waiting-time and transition-based estimators form a broad programme [3, 4, 8, 9, 10, 11, 12, 13]. These methods motivate an exact test: can a larger microscopic model dissipate less than *every* smallest model while preserving the complete observed law? A numerical local minimum within one topology cannot answer this question.

Hidden states also matter in other physical implementation problems. Owen, Kolchinsky and Wolpert study how many hidden states are needed to implement a prescribed conditional distribution with potentially time-dependent master equations, and how this requirement interacts with other resources [18]. The constraint here is an entire stationary path law generated by a time-homogeneous chain. Preserving that law fixes all observed temporal statistics, rather than a single input–output transition map. In control theory and queueing theory it is classical that a positive (nonnegative-rate) realisation of a transfer function may need more states than its minimal linear realisation [15, 16]. Our situation is sharper: a positive realisation with the minimal number of states exists, yet a larger one is thermodynamically cheaper. The new point is not that a mode can cancel from an observed transform, which is classical, but that an admissible cancelled mode can strictly lower the nonlinear flux functional that defines entropy production.

We give an analytic affirmative answer in continuous time. Our signal has two visible labels. Visits alternate between them, and their complete durations are mutually independent. Such an observed stationary process is invariant under time reversal: reversing the order of independent complete visits and interchanging the age and residual life of the visit containing the origin preserves its law. Nevertheless, its microscopic Markov realisations can be forced to dissipate. Waiting-time information can reveal this hidden nonequilibrium even when the observed arrow of time vanishes [3, 4].

Relation to the two closest sources. The family used here was specified in Aznagulov's exact-law architecture problem [5]. That work proves that the fixed-dimension cost V_n is *attained* for every n, introduces the architecture penalty $G_\kappa(P) = \min_n[V_n(P) + \kappa n]$, gives complete phase-type cancellation equations for this family with a four-phase example showing why certain restricted parametrisations miss valid models, proves positive minimal cost at every finite dimension, and asks whether growing architecture can drive the cost to its all-dimension infimum, possibly zero. In its discussion, profitable insertion of hidden phases is raised as a possible route, not established. An Evidence Press research candidate establishes a zero-cost classification for alternating-renewal observations and a positive architecture-independent floor, $V_\infty(P_{1/20}) > 0.017568198744$ [6]. Relative to these sources, the present paper proves: the exact three-state minimum and its unique minimiser; a strict four-state improvement for every q, so that by [5] the attained minimum $V_4(P_q)$ lies strictly below $V_3(P_q)$ and the optimal architecture in G_κ has more than three states once $\kappa < V_3 - V_4$; and a closed-form architecture-independent floor $(2/3)(1-q)\log(1/q)$, which at $q = 1/20$ is about 108 times the floor certified in [6]. It

Table 1: What is fixed, over which architectures, and what is proved. "Exact law" means the complete stationary observed path law.

Source	Constraint	Architecture	Result relevant here
Ehrich [1]	all observed statistics	known hidden size	tightest bound at that size
Skinner–Dunkel [3, 4]	waiting-time data	optimised topologies	lower bounds; flux merging
Aznagulov [5]	exact law	each fixed n	V_n attained; $V_n(P_q) > 0$
EP floor [6]	exact law	all finite n	$V_\infty(P_{1/20}) > 0.0176$
This paper, Thm 1	exact law	$n = 3$ vs $n = 4$	$V_4(P_q) < V_3(P_q)$ for all q
This paper, Thm 2, 5	(r, m, f_0) only	all finite n	sharp, unattained; $V_\infty(P_{1/20}) \geq 1.897$

does not settle whether $V_\infty(P_q)$ is attained. Table 1 summarises the comparison.

The second contribution is an exact optimisation involving only three observable dwell statistics. It supplies a lower bound valid for arbitrary hidden dimension and arbitrary rates, including observations with correlated successive visits. Its positive infimum is not attained, even though a sequence with only three states approaches it. Here the full observed law changes along the sequence. We identify the lost short-time constraint in its limit, which explains why this statement does not resolve nonattainment under a fixed complete law.

The mathematical ingredients include stationary-flux contraction, phase-type representations and inequalities between symmetric divergences. Flux contraction and optimisation from partial statistics have established precedents [3, 4]; the information inequalities also belong to an established literature [7]. We give the necessary proofs with explicit normalisations. Our claims concern the exact optimisation problems and the architecture comparison, rather than priority for these ingredients.

Roadmap. Section 2 fixes notation, the cost functional and the target signal. Section 3 proves the three-state minimum, verifies the explicit four-state competitor and extends the improvement to every q. Section 4 proves the architecture-independent lower bound, and Section 5 shows that it is the exact answer when only three statistics are fixed. Section 6 reports numerical exploration of the remaining gap, clearly separated from the proofs. Section 7 interprets the results for inference. Readers who want only the main message can read this introduction, Theorem 1, Figure 1 and Section 7.

2 Models, observations and optimisation

Let Q be a finite irreducible continuous-time Markov generator in row convention, with stationary probability row vector π. A deterministic map labels each microscopic state A or B, and both labels are used. The chain starts in stationarity. All microscopic variables are even under ordinary time reversal. With natural logarithms and Boltzmann's constant one, define

$$F_{ij} = \pi_i Q_{ij}, \qquad \sigma(Q) = \sum_{i<j} J(F_{ij}, F_{ji}), \qquad J(x, y) = (x - y)\log(x/y). \tag{1}$$

We set $J(0, 0) = 0$ and $J(x, 0) = J(0, x) = +\infty$ for $x > 0$. The cost has units of inverse time. Its interpretation as entropy production assumes the microscopic Markov description and reversal convention stated above; no inference about odd variables or non-Markovian microscopic dynamics is made.

For a complete stationary observed path law P, write

$$V_n(P) = \inf\{\sigma(Q) : |Q| \le n,\ P_{Q,h} = P\}, \qquad V_\infty(P) = \inf_{n\ge 1} V_n(P). \tag{2}$$

Equality of path laws includes all visit durations and all dependencies between visits, with their stationary initialisation. It is stronger than agreement of finitely many moments, correlation functions or sampled transition probabilities. Every exact competitor supplies an *upper* bound on V_∞. A minimum within a restricted architecture supplies a lower bound on the physical entropy only if that architecture restriction is independently justified.

For $0 < q < 1$, set

$$s = 1 + q, \qquad b = \frac{2q}{s}, \qquad c = 1 + q + q^2, \qquad D = 1 - q. \tag{3}$$

The target P_q is the stationary alternating-renewal law with mutually independent complete visits, A duration $\mathrm{Exp}(s)$, and B duration density f_q whose Laplace transform is

$$\widehat{f}_q(z) = \frac{bz + c}{z^2 + 2sz + c}, \qquad z \ge 0. \tag{4}$$

It has mean $m_B = 2/s$, initial density $f_q(0) = b$, stationary B mass $2/3$, and entrance intensity $r = s/3$ into either label. The following ring is an exact realisation:

$$Q_q = \begin{pmatrix} -s & 1 & q \\ q & -s & 1 \\ 1 & q & -s \end{pmatrix}, \qquad \pi_q = (1/3, 1/3, 1/3), \tag{5}$$

with the first state labelled A. Its cost is

$$\sigma(Q_q) = D \log(1/q). \tag{6}$$

The rate one in this definition fixes the reference time unit. Multiplying every rate by $\gamma > 0$ rescales time and multiplies every displayed cost and entrance intensity by γ; $mf(0)$ remains unchanged.

A single A state gives the useful phase-type form

$$Q = \begin{pmatrix} -s & s\alpha \\ t & S \end{pmatrix}, \qquad t = -S\mathbf{1}, \quad \alpha\mathbf{1} = 1. \tag{7}$$

Here S is a transient Metzler matrix: its off-diagonal entries are nonnegative, its row sums are nonpositive, and absorption occurs with probability one. The row α is the distribution on entering B. The complete B density at age θ is $\alpha e^{\theta S} t$. Define the occupation row per visit $\mu = \alpha(-S)^{-1}$. For an exact realisation of P_q,

$$\mu\mathbf{1} = 2/s, \quad \mu t = 1, \quad \pi = (1/3, r\mu), \quad r = s/3. \tag{8}$$

The entropy formula becomes

$$\sigma(Q) = r\left\{\sum_i J(\alpha_i, \mu_i t_i) + \sum_{i<j} J(\mu_i S_{ij}, \mu_j S_{ji})\right\}. \tag{9}$$

3 Exact thermodynamic advantage of a fourth state

Theorem 1 (Strict benefit of hidden architecture)**.** *For every $0 < q < 1$, the least number of microscopic states realising P_q is three, and*

$$V_3(P_q) = (1-q)\log(1/q). \tag{10}$$

The unique three-state minimiser, up to exchange of the B states, is (5). The same lower bound $(1-q)\log(1/q)$ holds for every exact realisation with at most two B states and any number of A states (Appendix A). For every $0 < q < 1$,

$$V_4(P_q) < V_3(P_q). \tag{11}$$

The improvement is achieved by a four-state exact realisation whose off-diagonal rates may all be chosen strictly positive; rational q admits rational rates. At $q = 1/20$, the following simpler finite-entropy generator, with one absent reciprocal pair of edges, gives a much larger explicit gap:

$$Q_* = \begin{pmatrix} -21/20 & 4781/4800 & 31/1200 & 9/320 \\ 1/20 & -79/60 & 19/15 & 0 \\ 4/5 & 1/3 & -131/60 & 21/20 \\ 21/20 & 0 & 1/5 & -5/4 \end{pmatrix} \tag{12}$$

with stationary distribution

$$\pi_* = (1/3, 217/720, 7/36, 41/240) \tag{13}$$

and only its first state labelled A has exactly the law $P_{1/20}$ and

$$V_4(P_{1/20}) \le \sigma(Q_*) < 2.774 < 2.845 < V_3(P_{1/20}). \tag{14}$$

In words: three states are the fewest that can produce the signal, the cheapest three-state mechanism is the symmetric driven ring, and adding one hidden state always permits a cheaper mechanism producing exactly the same signal. The reason this is possible is that the signal fixes only the observed dwell law, a rational function, while the cost depends on every individual microscopic current; an extra hidden mode that cancels from the dwell law leaves room to redistribute those currents more cheaply. By the fixed-dimension attainment theorem of [5], $V_4(P_q)$

is itself attained, so (11) compares two attained minima. Figure 1 draws both mechanisms at $q = 1/20$. The comparison is between an explicit four-state cost and a *global* three-state minimum. It is not a comparison of two local numerical optima. The zeros in (12) occur as a reciprocal pair; every present edge has a positive reverse, so its cost is finite. We prove the global minimum, verify the rational competitor, and then give the full-family construction.

3.1 All two-phase realisations and their minimum

The denominator in (4) factors as

$$p(z) = z^2 + 2sz + c = (z+\ell)(z+H), \qquad \ell = s - \sqrt{q}, \quad H = s + \sqrt{q}. \tag{15}$$

The numerator at the two poles is nonzero: $c - b\ell = \ell(H-b) > 0$ and $c - bH = H(\ell - b) > 0$. Thus $\widehat{f}_q$ has reduced degree two. In particular B cannot consist of one microscopic state. Two total states are insufficient; with three total states the only possible label allocation is one A and two B states.

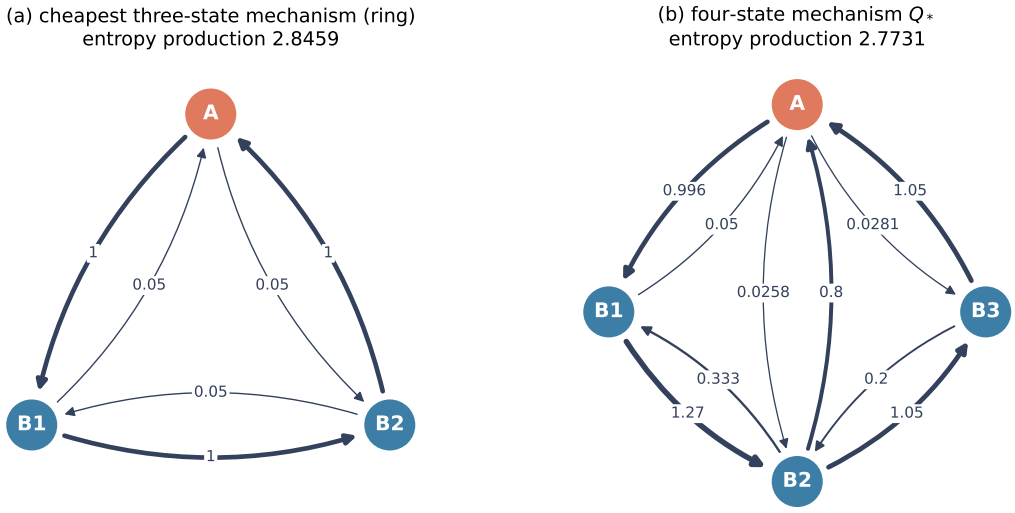


Figure 1: Two microscopic mechanisms for the same observed signal at $q = 1/20$. Only the label (A or B) of the current state is observed. (a) The three-state ring (5), the unique cheapest three-state mechanism. (b) The four-state generator (12). Its observed signal is identical in every statistical respect, yet it dissipates about 2.6% less. The pair of edges between $B1$ and $B3$ is absent in both directions.

Write the two-phase realisation as

$$S = \begin{pmatrix} -(x+u) & u \\ v & -(y+v) \end{pmatrix}, \quad t = (x, y)^{\mathsf{T}}, \quad \alpha = (a_0, 1-a_0). \tag{16}$$

Equal exits would give an exponential dwell and are excluded. Relabel to make $x < y$. Minimality of the degree-two transfer fixes the characteristic polynomial of S to p. Matching trace, determinant and initial density gives

$$u = \frac{x^2 - 2sx + c}{y - x}, \quad v = -\frac{y^2 - 2sy + c}{y - x}, \quad \alpha = \left(\frac{y-b}{y-x}, \frac{b-x}{y-x}\right). \tag{17}$$

Since $0 < b < \ell < H$, the complete nonnegative parameter domain is

$$0 \le x \le b, \qquad \ell \le y \le H. \tag{18}$$

Indeed $\alpha t = b$ places $x \le b \le y$; then $u > 0$, while $v \ge 0$ places y between the two roots. Conversely (17) on this rectangle gives a transient subgenerator and the target transform.

Every boundary face has infinite entropy. If $x = 0$, entry into the first B state has no direct reverse; if $x = b$, the second state exits with positive flux but has zero entrance flux. At $y = \ell$ or $y = H$, one internal edge is unidirectional. All states remain used with positive stationary mass. We may therefore minimise on the open rectangle.

Direct inversion gives

$$\mu = \left(\frac{2y/s - 1}{y - x}, \frac{1 - 2x/s}{y - x}\right). \tag{19}$$

Stationarity gives a common oriented cycle current per visit,

$$j = \alpha_1 - \mu_1 x = \mu_1 u - \mu_2 v = \mu_2 y - \alpha_2 = \frac{x + y - b - 2xy/s}{y - x} > 0. \tag{20}$$

Thus

$$\sigma = \frac{s}{3} j\mathcal{A}, \qquad \mathcal{A} = \log \frac{\alpha_1 uy}{\alpha_2 vx}. \tag{21}$$

The following change of variables makes the global inequality elementary:

$$a = s - 2x, \quad h = 2y - s, \quad R = \frac{a - D}{a + D}, \quad K = \frac{h - D}{h + D}. \tag{22}$$

The open domain is $-q < R < q$ and $-\sqrt{q} < K < \sqrt{q}$. Substitution in (21) yields

$$\sigma = \frac{D}{3} \frac{1 + RK}{1 - RK} \log \frac{(1 - qR^2)(1 - q^2K^2)}{(q^2 - R^2)(q - K^2)}. \tag{23}$$

In particular $sj = (D^2 + ah)/(a + h) = D(1 + RK)/(1 - RK) > 0$, which also proves the sign asserted in (20).

If $RK \geq 0$, the current factor is at least one. Each of the two logarithmic terms is increasing in the relevant square, so their sum is at least $3 \log(1/q)$. Equality requires $R = K = 0$.

If $RK < 0$, put

$$u_0 = |R|/q, \quad v_0 = |K|/\sqrt{q}, \quad \tau = q^{3/2}, \quad w = u_0 v_0 \in (0, 1). \tag{24}$$

The affinity is

$$\mathcal{A} = -2 \log \tau + \log \frac{1 - \tau^2 u_0^2}{1 - u_0^2} + \log \frac{1 - \tau^2 v_0^2}{1 - v_0^2}. \tag{25}$$

Expand each last term as $\sum_{n \geq 1} (1 - \tau^{2n}) u_0^{2n}/n$, or its v_0 counterpart. The coefficients are positive. The arithmetic–geometric mean inequality at each power gives

$$\mathcal{A} \geq 2 \log \frac{1 - \tau^2 w}{\tau(1 - w)}. \tag{26}$$

Let $z = \tau w$, $C = (1 - \tau)/(1 + \tau)$ and $T = (1 + z)/(1 - z) > 1$. Then $0 < C < CT < 1$. Multiplying (26) by the current factor gives

$$\begin{aligned} \frac{1 - z}{1 + z} 2 \log \frac{1 - \tau z}{\tau - z} &= \frac{4}{T} \operatorname{artanh}(CT) \\ &> 4 \operatorname{artanh} C = 2 \log(1/\tau) = 3 \log(1/q). \end{aligned} \tag{27}$$

The strict inequality follows from strict increase of $\operatorname{artanh}(v)/v$ on $(0, 1)$, also immediate from its positive power series. Equation (23) proves the global minimum. Equality is possible only at $R = K = 0$, which gives $x = q$, $y = 1$, $u = 1$, $v = q$ and $\alpha = (1/s, q/s)$. This is precisely the ring (5). Its reverse is obtained by exchanging the two B states.

3.2 Exact four-state realisation and certified gap

For (12), the B occupation, entrance and exit vectors are

$$\mu = (31/36, 5/9, 41/84), \quad \alpha = (683/720, 31/1260, 3/112), \quad t = (1/20, 4/5, 21/20)^{\mathsf{T}}. \tag{28}$$

All entries are positive. The B submatrix S_* satisfies the exact identities

$$S_* \mathbf{1} = -t, \quad \mu S_* = -\alpha, \quad \mu \mathbf{1} = 40/21, \quad \mu t = 1, \quad \alpha t = 2/21, \tag{29}$$

and

$$\left(S_*^2 + \frac{21}{10} S_* + \frac{421}{400} I \right) \mathbf{1} = 0. \tag{30}$$

These identities verify stationarity and full-law matching. In detail, the survival function $g(\theta) = \alpha e^{\theta S_*}\mathbf{1}$ obeys $g'' + (21/10)g' + (421/400)g = 0$, $g(0) = 1$ and $g'(0) = -2/21$. Its density transform is therefore exactly (4) at $q = 1/20$. Every return to the single A state resets the entrance distribution, independently of previous visits. Its holding rate is $21/20$. Consequently all complete visits have the required independent laws. For a stationary alternating-renewal process, the segment containing time zero has the standard equilibrium age and residual-life law, which is determined by the complete dwell laws and the stationary label probabilities; later segments are independent complete visits. The stationary fractions agree, so the entire stationary observation process agrees.

The additional hidden mode is explicit:

$$\det(zI - S_*) = (z + 53/20)\left(z^2 + \frac{21}{10}z + \frac{421}{400}\right). \tag{31}$$

The factor $z + 53/20$ cancels from the observed dwell transform. This is an exact cancellation, rather than a mode made difficult to observe by a small amplitude. It can nevertheless alter microscopic currents and their entropy cost.

The five present unoriented edges give

$$\begin{aligned}\sigma(Q_*) = {} & \frac{1141}{3600}\log\frac{683}{31} + \frac{529}{3600}\log\frac{560}{31} + \frac{17}{100}\log\frac{287}{15} \\ & + \frac{1141}{3600}\log\frac{589}{100} + \frac{17}{100}\log\frac{245}{41}.\end{aligned} \tag{32}$$

Rational upper and lower bounds for the logarithms prove

$$2.773076657 < \sigma(Q_*) < 2.773076658, \qquad \frac{19}{20}\log 20 - \sigma(Q_*) > 0.072869002. \tag{33}$$

The certified gap exceeds 2.56% of the three-state minimum. The percentage describes this competitor, not the maximum possible saving. Appendix C states the certificate method. This completes the pointwise comparison in Theorem 1.

3.3 A strict improvement throughout the family

Fix q and write $z_0 = s/2$, $\nu = D/s$. Insert a third B phase with exit rate z_0, between the original exit rates q and 1. Order the phases as x, y, w, with

$$t = (q, 1, z_0)^{\mathsf{T}}, \qquad \mu = (\rho, \rho, \varepsilon), \qquad \rho = 1/s - \varepsilon/2. \tag{34}$$

For a finite parameter $M \geq 2$, define

$$A_0 = M/2, \quad B_0 = sM/4, \quad C_0 = qM/2 + \nu, \quad E_0 = sM/4 + D/2. \tag{35}$$

The internal fluxes per visit are

$$\begin{aligned}&L_{xw} = \varepsilon A_0, \quad L_{wx} = \varepsilon B_0, \quad L_{yw} = \varepsilon C_0, \quad L_{wy} = \varepsilon E_0, \\ &L_{xy} = \rho - \varepsilon A_0/2, \qquad L_{yx} = q\rho - \varepsilon C_0/2.\end{aligned} \tag{36}$$

Put $S_{ij} = L_{ij}/\mu_i$ for $i \neq j$ and $S_{ii} = -t_i - \sum_{j\neq i} S_{ij}$. The entrance row is

$$\alpha = \left(\frac{1 - q\varepsilon}{s}, \frac{q(1-\varepsilon)}{s}, \frac{2q\varepsilon}{s}\right). \tag{37}$$

For fixed M, sufficiently small positive ε makes every off-diagonal rate and every entrance positive. The positive exits ensure transience, and (7) is irreducible.

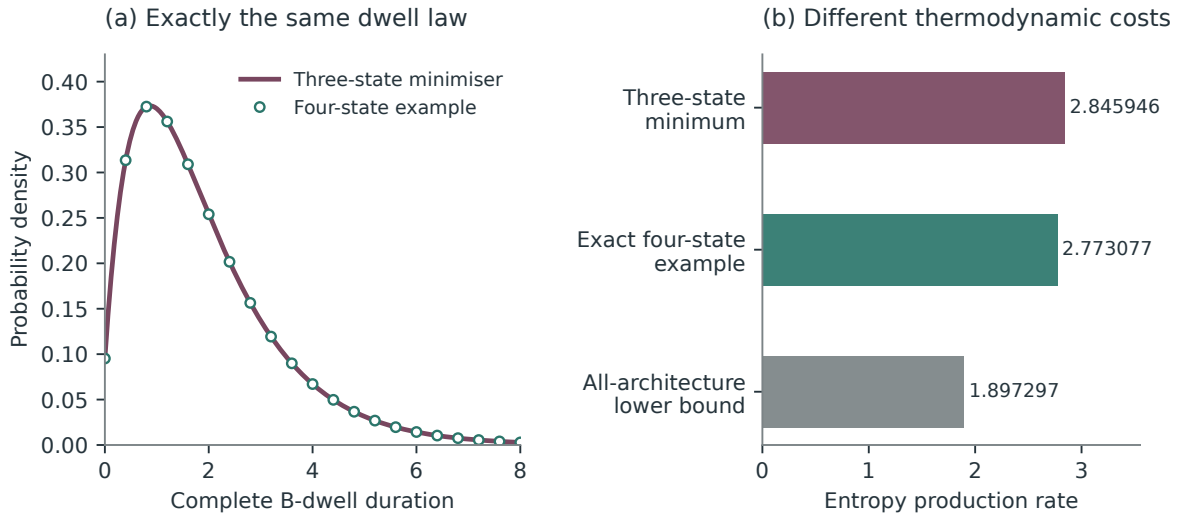


Figure 2: Exact observation and different thermodynamic costs at $q = 1/20$. (a) Complete B-dwell densities calculated independently from the three-state and four-state generators coincide; equality is proved for the entire stationary process, beyond the plotted marginal. (b) The four-state example lies strictly below the proved three-state minimum. The universal lower bound from Corollary 4 applies to all finite architectures; its bar does not represent an attaining model. Rates use the reference time unit of (5).

All these admissible models reproduce P_q exactly. To see this, let $F(v) = v^2 - 2sv + c$. Then

$$F(q) = D, \quad F(1) = -qD, \quad F(z_0) = D^2/4. \tag{38}$$

The fluxes satisfy

$$\sum_{j \neq i} L_{ij}(t_j - t_i) = \mu_i F(t_i). \tag{39}$$

At x and y the terms proportional to A_0 and C_0 cancel; at w the identity follows from $E_0 - B_0 = D/2$. Thus $St = c\mathbf{1} - 2st$ and $p(S)\mathbf{1} = 0$. Directly, $\mu S = -\alpha$, $\mu\mathbf{1} = 2/s$, $\mu t = 1$, $\alpha\mathbf{1} = 1$ and $\alpha t = b$. The survival-function argument used for (30), together with reset independence, proves full-law equality.

It remains to prove a strict entropy decrease. Let $\Psi_M(\varepsilon)$ be the entropy per visit, the expression in braces in (9). As $\varepsilon \downarrow 0$, the new phase disappears and the remaining two phases are the optimal ring. The three new unoriented edges contribute exactly linear terms in ε, by homogeneity of J. The old edges are smooth near zero. Put $\Lambda = \log(1/q)$ and $h_+ = \Lambda + D$, $h_- = -\Lambda - D/q$. Their derivatives and the new-edge terms give

$$\begin{aligned} d_M := \Psi'_M(0+) = {} & h_+(-1 - q/s - M/4) + h_-(-q - 1/2 - qM/4) \\ & + J(b, z_0) + J(M/2, sM/4) + J(qM/2 + \nu, sM/4 + D/2). \end{aligned} \tag{40}$$

The terms linear in M cancel. Expanding the last J term about $(qM/2, z_0M/2)$ gives the finite limit

$$\begin{aligned} d_\infty := \lim_{M\to\infty} d_M &= \frac{D^2}{2s}\left[3 + \log(s/2) - \frac{1+2q}{D}\log(1/q)\right] \\ &= \frac{\nu^2}{1+\nu}\left[3 - \frac{1}{2}\log(1-\nu^2) - \frac{3\,\operatorname{artanh}\nu}{\nu}\right]. \end{aligned} \tag{41}$$

Its sign is exact, since for $0 < \nu < 1$ the bracket equals

$$\sum_{n=1}^{\infty} \frac{1-4n}{2n(2n+1)} \nu^{2n} < 0. \tag{42}$$

Choose a finite M with $d_M < 0$, then choose a positive admissible ε sufficiently small that $\Psi_M(\varepsilon) < \Psi_M(0)$. This proves (11). Appendix B supplies rational formulas for both parameters and a quantified strict decrease, eliminating any need for a numerical search in the construction.

The additional decay rate is

$$\kappa = 1 + sM/2 + \frac{\varepsilon(sM/2+\nu)}{2\rho}, \tag{43}$$

and $\det(\lambda I - S) = (\lambda + \kappa)p(\lambda)$. Indeed the target's two poles must be eigenvalues of S and the remaining root follows from its trace. Since $M \geq 2$ gives $\kappa > H$, this mode is distinct and cancels from the observed law. The construction changes entropy through an additional mode, rather than by duplicating a state with unchanged flux ratios.

4 An architecture-independent lower bound from short dwells

The exact-law optimum remains below the three-state value. To bound it from below without restricting hidden dimension, we use the initial density of a complete visit. The next theorem applies to any stationary binary partition; successive visits may be dependent.

Theorem 2 (Suppressed immediate returns)**.** *Let $r > 0$ be the stationary entrance intensity into B, let m be the mean complete B dwell under sampling at entrances, and let $f_0 = f_B(0)$ be its right-hand density at zero. If $0 < k = mf_0 < 1$, every finite realisation satisfies*

$$\sigma(Q) > 4r\sqrt{1-k}\,\operatorname{artanh}\sqrt{1-k}. \tag{44}$$

If $f_0 = 0$, every compatible finite realisation has infinite entropy production.

To prove the theorem, define for every microscopic B state

$$t_i = \sum_{a\in A} Q_{ia}, \quad \mu_i = \frac{\pi_i}{r}, \quad \alpha_i = \frac{\sum_{a\in A}\pi_a Q_{ai}}{r}, \quad \beta_i = \mu_i t_i. \tag{45}$$

Here α and β are the entrance and exit distributions at stationary crossings. Both sum to one. Stationary occupation per entrance gives $\mu\mathbf{1} = m = \pi(B)/r$, and the killed-chain formula gives $f_0 = \alpha t$. None of these identities requires renewal independence.

Let $\mathcal{J}(p,q) = \sum_i (p_i - q_i)\log(p_i/q_i)$ denote the sum, rather than the average, of the two relative entropies. Applying log-sum to the boundary edges at each B state and discarding nonnegative internal contributions gives

$$\sigma(Q) \geq r\mathcal{J}(\alpha,\beta). \tag{46}$$

The Bhattacharyya affinity is bounded by Cauchy–Schwarz:

$$a = \sum_i \sqrt{\alpha_i\beta_i} = \sum_i \sqrt{\mu_i}\sqrt{\alpha_i t_i} \leq \sqrt{mf_0}. \tag{47}$$

Sason's related sharp comparison fixes total variation and defines Jeffreys divergence as one half of $\mathcal{J}$ [7]. We require an affinity-constrained inequality instead, and prove it here without inferring it from that total-variation comparison.

The function $4d\operatorname{artanh} d$ appearing below is the entropy production of a single biased link carrying fluxes proportional to $(1 \pm d)/2$; functions of this form also control optimal thermodynamic uncertainty relations [14].

Lemma 3 (Affinity and symmetric relative entropy)**.** *For probability vectors p, q with affinity $a = \sum_i \sqrt{p_i q_i}$,*

$$\mathcal{J}(p,q) \geq \Phi(a), \qquad \Phi(a) = 4\sqrt{1-a^2}\,\operatorname{artanh}\sqrt{1-a^2}. \tag{48}$$

The inequality is sharp for each $0 < a \leq 1$.

Proof. It suffices to consider finite divergence. Put $w_i = (p_i + q_i)/2$, $u_i = (p_i - q_i)/(p_i + q_i)$ and $z_i = \sqrt{1-u_i^2}$ on indices of positive total mass. Then $\sum_i w_i = 1$, $a = \sum_i w_i z_i$ and $\mathcal{J}(p,q) = \sum_i w_i \Phi(z_i)$. For $d = \sqrt{1-z^2}$,

$$\Phi'(z) = -4\left(\frac{z}{d}\operatorname{artanh} d + \frac{1}{z}\right) < 0, \qquad \Phi''(z) = \frac{4}{d^3}\left(\frac{d}{1-d^2} - \operatorname{artanh} d\right) > 0. \tag{49}$$

The bracket has value zero at $d = 0$ and positive derivative $2d^2/(1-d^2)^2$. Jensen's inequality proves (48); endpoint cases follow by limits. Equality holds for $p = ((1+d)/2, (1-d)/2)$ and $q = ((1-d)/2, (1+d)/2)$. □

Equations (46)–(48), and decrease of Φ, give the non-strict form of (44). Equality would require zero entropy on all B-internal edges. Their stationary fluxes would be symmetric, so microscopic stationarity would give $\alpha = \beta$. Then

$$1 = (\mu t)^2 \leq (\mu \mathbf{1}) \sum_i \mu_i t_i^2 = m f_0, \tag{50}$$

a contradiction. Infinite entropy also satisfies the strict inequality. Finally, finite entropy requires every used boundary edge to have a reverse; at least one such edge exists. For its B endpoint both α_i and t_i are positive, forcing $f_0 > 0$. This proves Theorem 2.

Corollary 4 (Unrestricted exact-law bracket)**.** *For $0 < q < 1$,*

$$\frac{2}{3}(1-q)\log(1/q) \leq V_\infty(P_q) \leq V_4(P_q) < V_3(P_q) = (1-q)\log(1/q). \tag{51}$$

At the explicit benchmark,

$$1.897297106584\ldots = \frac{19}{30}\log 20 \leq V_\infty(P_{1/20}) \leq \sigma(Q_*) = 2.773076657835\ldots. \tag{52}$$

Proof. The observed statistics are $r = s/3$, $m = 2/s$ and $f_0 = 2q/s$. Hence $\sqrt{1-mf_0} = D/s$ and $\operatorname{artanh}(D/s) = \frac{1}{2}\log(1/q)$. Apply Theorem 2 to every exact realisation, then take the infimum. The explicit four-state model supplies the upper bound. □

The strict inequality for each finite model does not justify replacing the lower $\leq$ in (52) by $<$ after taking the all-dimension infimum. The endpoints are certified bounds, not confidence limits or estimates of an optimum.

Equation (51) also bounds the benefit of unrestricted architecture: its relative saving from the minimum-state cost is positive and at most one third. It fixes both endpoint orders,

$$V_\infty(P_q) = \Theta((1-q)^2) \quad (q \uparrow 1), \qquad V_\infty(P_q) = \Theta(\log(1/q)) \quad (q \downarrow 0). \tag{53}$$

These are order statements, not determinations of the leading coefficients.

5 Exact optimisation for three observable statistics

The lower bound has a complete optimality statement when the observed constraints are precisely (r, m, f_0). It is useful to solve that problem separately, because it shows exactly what these statistics can and cannot force thermodynamically.

Theorem 5 (Sharp cost and nonattainment)**.** *Fix $r, m, f_0 > 0$ with $0 < rm < 1$. Over all finite irreducible stationary binary-labelled chains with these three B statistics,*

$$\inf \sigma = \begin{cases} 4r\sqrt{1-mf_0}\,\operatorname{artanh}\sqrt{1-mf_0}, & 0 < mf_0 < 1, \\ 0, & mf_0 \geq 1. \end{cases} \tag{54}$$

The positive infimum is attained by no finite model, but is approached by three-state models. The zero infimum is attained by a reversible model with at most three states.

5.1 Positive branch and a fixed-dimension limiting sequence

Let $d = \sqrt{1-mf_0} \in (0,1)$ and choose $M > d/2$. Set

$$\mu = (m/2, m/2), \quad \alpha = ((1+d)/2, (1-d)/2), \quad t = ((1-d)/m, (1+d)/m)^{\mathsf{T}}, \tag{55}$$

$$S_M = \frac{1}{m}\begin{pmatrix} -(1+2M) & 2M+d \\ 2M-d & -(1+2M) \end{pmatrix}, \qquad s_A = \frac{r}{1-rm}, \quad Q_M = \begin{pmatrix} -s_A & s_A\alpha \\ t & S_M \end{pmatrix}. \tag{56}$$

All three unoriented edges have positive rates in both directions. The identities $-S_M\mathbf{1} = t$ and $\mu S_M = -\alpha$ give

$$\pi_M = (1 - rm, rm/2, rm/2). \tag{57}$$

The entrance intensity, mean and initial density are respectively r, m and $(1-d^2)/m = f_0$, exactly for every M. Boundary fluxes per visit are α and its reversed vector; the two internal fluxes are $M + d/2$ and $M - d/2$. Therefore

$$\sigma(Q_M) = r\left\{4d\operatorname{artanh} d + d\log\frac{M+d/2}{M-d/2}\right\}. \tag{58}$$

The last term tends to zero as $M \to \infty$. Theorem 2 proves that the limit is the infimum and that no finite model attains it. Nonattainment is thus immediate from the strict inequality of Theorem 2; the new content of Theorem 5 is that the bound is sharp, and that three states suffice to approach it.

The full dwell law is not constant in this sequence. Direct inversion gives

$$\widehat{f}_M(z) = \frac{(1-d^2)mz + (1+4M+d^2)}{m^2z^2 + 2m(1+2M)z + (1+4M+d^2)} \longrightarrow \frac{1}{1+mz}. \tag{59}$$

The limit is an exponential dwell of mean m, whose density at zero is $1/m$. Every finite member has initial density $(1-d^2)/m < 1/m$. The fast internal mode confines the initial-density discrepancy to a shrinking time interval. Evaluation of $f(0)$ consequently fails to commute with this limit. A compactness result for a fixed complete law cannot be applied to the present constraint set, which is not closed under this convergence.

For the target statistics of P_q, the original ring occurs at $M = 1/2$ in (56). Raising M lowers its cost while preserving (r, m, f_0), but changes its dwell transform. It is therefore not an exact-law competitor. This supplies a direct check on the distinction between Theorems 1 and 5.

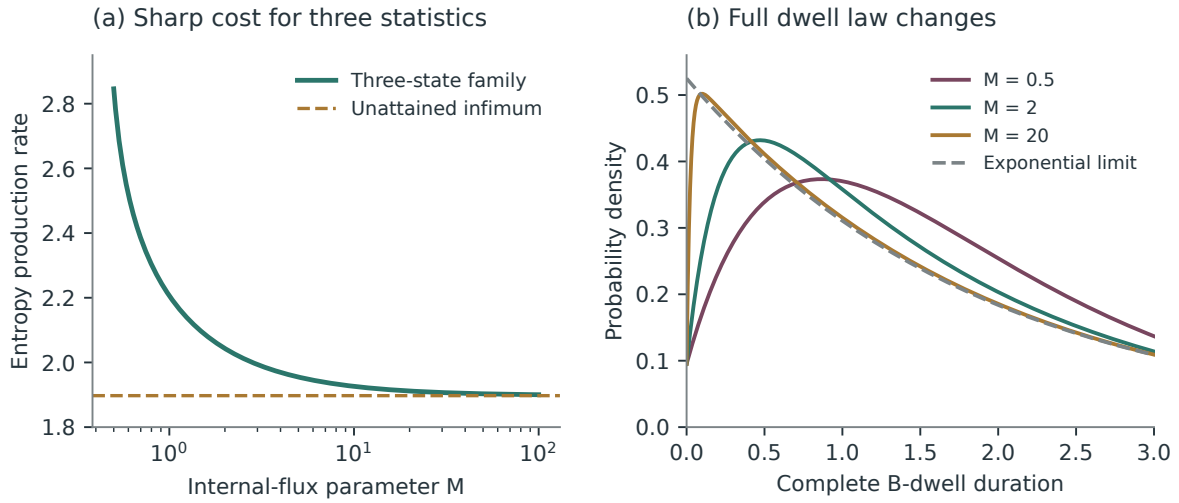


Figure 3: The statistics-only optimum has different behaviour from the exact-law architecture problem. (a) For the (r, m, f_0) of $P_{1/20}$, (58) approaches its sharp positive infimum as M grows. (b) The complete dwell density changes along this family. Its limiting exponential has a different value at zero despite equality of f_0 for every finite M. Thus this sequence cannot establish fixed-law nonattainment. The time and rate units are those of Figure 2.

5.2 Zero branch

When $k = mf_0 = 1$, a two-state chain with rates $s_A = r/(1 - rm)$ and $1/m$ is reversible and has the required statistics. When $k > 1$, put

$$w_2 = \frac{1}{4k-3}, \quad w_1 = 1 - w_2, \quad \mu = m(w_1, w_2), \quad t = \left(\frac{1}{2m}, \frac{2k-1}{m}\right)^{\mathsf{T}}. \tag{60}$$

Let $S = -\operatorname{diag}(t)$ and $\alpha_i = \mu_i t_i$, using the same reset rate s_A. One checks $\alpha\mathbf{1} = 1$, $\mu\mathbf{1} = m$ and $\alpha t = f_0$. Every boundary stationary flux equals its reverse; there are no internal B edges. The resulting three-state star is irreducible and reversible. This proves the remaining part of Theorem 5.

6 How large can the architecture saving be?

The theorems above leave a gap at $q = 1/20$ between the architecture-independent floor 1.8973 and the four-state cost 2.7731. This section narrows the upper end with a further exact competitor and reports numerical exploration of the remaining questions. Only Proposition 6 is a proved statement; the rest is discovery evidence obtained by local optimisation from many random starts, and it is not used in any proof.

Proposition 6 (Improved exact four- and five-state competitors)**.** *At $q = 1/20$ the rational generator, with the first state labelled A,*

$$Q_\sharp = \begin{pmatrix} -\frac{21}{20} & \frac{483}{20000} & \frac{974299}{28880000} & \frac{28652249}{28880000} \\ \frac{134}{125} & -\frac{93181}{75375} & \frac{12379}{75375} & 0 \\ \frac{1541}{2000} & \frac{1869}{2000} & -\frac{2969283}{1444000} & \frac{507263}{1444000} \\ \frac{97}{2000} & 0 & \frac{3812009}{2888000} & -\frac{3952077}{2888000} \end{pmatrix} \tag{61}$$

realises $P_{1/20}$ exactly and has

$$\sigma(Q_\sharp) = 2.765997710528\ldots, \qquad V_3(P_{1/20}) - \sigma(Q_\sharp) > 0.0799479, \tag{62}$$

a saving of 2.81%. *The rational five-state generator*

$$Q_5 = \begin{pmatrix} -\frac{21}{20} & \frac{1113}{80000} & \frac{1701}{80000} & \frac{50952901}{51480000} & \frac{129029}{5148000} \\ \frac{139}{125} & -\frac{63247}{46500} & \frac{11539}{46500} & 0 & 0 \\ \frac{463}{500} & \frac{897}{1000} & -\frac{1257919}{469000} & 0 & \frac{100733}{117250} \\ \frac{6}{125} & 0 & 0 & -\frac{246223}{160875} & \frac{238501}{160875} \\ \frac{1383}{2000} & 0 & \frac{2323}{1000} & \frac{33907}{46800} & -\frac{54683}{14625} \end{pmatrix} \tag{63}$$

also realises $P_{1/20}$ *exactly, with*

$$\sigma(Q_5) = 2.746985402969\ldots, \qquad V_3(P_{1/20}) - \sigma(Q_5) > 0.0989602, \tag{64}$$

a saving of 3.48%. *Consequently* $1.8973 \le V_\infty(P_{1/20}) \le V_5(P_{1/20}) \le 2.7469855$ *and* $V_4(P_{1/20}) \le 2.7659978$.

Proof. The verification follows Section 3 line by line, in exact rational arithmetic. The rows sum to zero, the off-diagonal entries are nonnegative and the only zeros form a reciprocal pair. With S the B block, $t = -S\mathbf{1}$, α the normalised first row and $\mu = \alpha(-S)^{-1}$, one has $\alpha\mathbf{1} = 1$, $\alpha t = 2/21$, $\mu\mathbf{1} = 40/21$, $\mu t = 1$, $\mu > 0$ and $(S^2 + \frac{21}{10}S + \frac{421}{400}I)\mathbf{1} = 0$. As for (30), these identities and the single reset state give the complete law $P_{1/20}$. The characteristic polynomial is $\det(zI - S) = (z + \frac{4459837153}{1741464000})(z^2 + \frac{21}{10}z + \frac{421}{400})$, so again one hidden mode cancels. The cost is a sum over five edges of terms $(x - y)\log(x/y)$ with rational x, y, enclosed by the series of Appendix C. The same identities hold for Q_5 with its four-state B block, whose zeros form three reciprocal pairs; there $\det(zI - S)$ is the product of

$z^2 + \frac{21}{10}z + \frac{421}{400}$ and $z^2 + \frac{134945227969}{18711693000}z + \frac{156924372390977}{14033769750000}$, so two hidden modes cancel, and the cost is a sum over seven edges. The stationary vector of $Q_\sharp$ is

$$\pi_\sharp = \Big(\frac{1}{3}, \frac{3650219496}{22299185765}, \frac{27820291057}{133795114590}, \frac{39475135027}{133795114590}\Big).$$

Both scripts print the exact stationary vectors and certify the rational enclosures

$$2.765997710527 < \sigma(Q_\sharp) < 2.765997710529, \qquad 2.746985402969 < \sigma(Q_5) < 2.746985402970.$$

The scripts `extension/verify_improved.py` and `extension/verify_five.py` perform every step with the Python standard library.

□

The four-state optimum. By the reset lemma (Appendix A), every four-state exact realisation can be replaced, at no greater cost, by one with a single A state and at most three B states. Those with at most two B states cost at least $V_3(P_{1/20})$, so a four-state minimiser below V_3 may be sought among models with one A and three B states. We searched this class from random starts, both over the stratum on which $p(S)\mathbf{1} = 0$, which contains Q_* and $Q_\sharp$ and is mapped to its reversed counterpart $\alpha\, p(S) = 0$ by time reversal, and, independently, over all three-phase representations constrained only by the first three derivatives $f(0), f'(0), f''(0)$ of the dwell density. The second problem is a relaxation of the exact-law problem, so its minimum cannot exceed $V_4(P_{1/20})$. Both searches returned the same best local value, 2.765996, with one reciprocal pair of internal edges tending to zero. $Q_\sharp$ is a rational rounding of the best local solution found, and its cost is within 2×10^{-6} of it. The coincidence of the two searches motivates, but does not certify, the conjecture that $V_4(P_{1/20}) = 2.76599\ldots$, attained on the boundary stratum on which one internal edge pair is absent. A certified global optimisation would be needed to turn this into a theorem. The searches used SLSQP with 150–400 random starts per stratum, enumerated every pattern of up to three absent edge pairs, and accepted only solutions with constraint residuals below 10^{-9}.

Table 2: Numerical exploration of the four-state optimum across the family. V_3 and the floor are proved; the third column is the best exact four-state cost found by local optimisation from random starts, so it is an upper estimate of V_4 and the saving is a lower estimate. For $q \geq 0.5$ the savings found are tiny and not monotone in q, so these rows are inconclusive about the size of the saving; Theorem 1 guarantees that every saving is positive.

q	V_3 (proved)	best four-state cost found	saving found	floor $\frac{2}{3}V_3$ (proved)
0.01	4.55912	4.29941	0.25970 (5.7%)	3.03941
0.05	2.84595	2.76600	0.07995 (2.8%)	1.89730
0.1	2.07233	2.03755	0.03477 (1.7%)	1.38155
0.2	1.28755	1.27774	0.00981 (0.76%)	0.85837
0.3	0.84278	0.83955	0.00323 (0.38%)	0.56185
0.5	0.34657	0.34657	4.7×10^{-6} (0.0013%)	0.23105
0.7	0.10700	0.10699	9.8×10^{-6} (0.0091%)	0.07133
0.9	0.01054	0.01054	6.4×10^{-8} (0.00061%)	0.00702

The whole family. Table 2 repeats the search for other values of q. Among the sampled values, the largest relative saving found, about 6%, is at the smallest $q = 0.01$ (strong driving), and the savings found become very small as q approaches one, where the ring approaches equilibrium. The data are consistent with, but do not establish, a vanishing relative saving as $q \uparrow 1$. The proved floor stays well below every four-state value, so the table does not indicate whether larger architectures could approach it.

Does a fifth state help? The same search with four hidden B states, over the stratum $p(S)\mathbf{1} = 0$, returned 2.746985, and the rational rounding Q_5 of Proposition 6 certifies a cost within 10^{-6} of this value. A search with five hidden B states, from random interior starts only, returned 2.739391. The successive best costs found,

$$2.845946\ (n = 3), \quad 2.765996\ (n = 4), \quad 2.746985\ (n = 5), \quad 2.739391\ (n = 6),$$

decrease with shrinking steps 0.080, 0.019 and 0.0076. Only the first is a proved minimum and the others are upper estimates, so this is not evidence of a limit. It does suggest that each added hidden state lowers the best achievable cost further, which bears on the question of [5] whether the selected state count diverges as the price per state tends to zero. If it does, no finite architecture attains $V_\infty(P_{1/20})$.

Can more local statistics sharpen the floor? A natural attempt to raise the lower bound adds further short-time statistics to (r, m, f_0). Numerical minimisation suggests that the first derivative $f'(0)$ does not help: with three hidden B states and the exact values of r, m, f_0 and $f'(0)$ for $P_{1/20}$, the best local cost found came within 3×10^{-6} of the floor 1.8973. If that local solution is close to the global one, $f'(0)$ alone cannot materially raise the floor. Matching $f''(0)$ as well raised the best three-hidden-state cost found to the four-state value 2.766, but runs with more hidden states did not reach the feasibility tolerance, so we draw no conclusion from them about larger architectures.

7 Interpretation, verification and limits

The exact architecture comparison has a direct inference consequence. Suppose the true physical generator is Q_*. An observer with its entire stationary binary signal can fit a three-state model perfectly. Minimising dissipation over every such three-state fit gives a value larger

than the physical dissipation of Q_*. Thus statistical adequacy and the fewest possible hidden states do not make that restricted minimum a valid architecture-independent lower bound. This conclusion concerns the exact-law optimisation. It does not invalidate lower bounds that optimise over a proved complete class or over a sufficiently broad relaxation of the observations.

The structural reason is visible in (30): a microscopic mode can cancel from the observed dwell law while changing the current distribution. Reducing a transfer function to its smallest linear realisation does not preserve the nonlinear flux functional (1). Nor does an arbitrary similarity transformation preserve nonnegative transition rates. The proof avoids both assumptions by checking all two-phase models analytically and constructing a positive larger model directly.

The short-dwell bound quantifies a complementary obstruction. If $mf_0 < 1$, the entrance and exit distributions of hidden states must differ. Their affinity is bounded away from one, forcing entropy at the observed boundary regardless of hidden dimension. The result detects a thermodynamic cost even without independent visits. It uses an infinitesimal property of a complete dwell density, however. Estimating f_0 from finite-resolution or finite-duration data requires further assumptions and analysis. In single-channel recording this is the classical missed-events problem: events shorter than the dead time are not detected and distort the apparent dwell distributions near zero [17]. We supply no statistical coverage statement and no experimental calibration. Units are fixed only by the time scale: if the rate-one transitions of (5) occurred at $10^3\,\mathrm{s}^{-1}$, every cost above would be multiplied by 10^3, so the explicit saving at $q = 1/20$ would be about $73\,k_B\,\mathrm{s}^{-1}$.

All principal claims are analytic. Computational exploration located the simple rational competitor on a branch satisfying $p(S)\mathbf{1} = 0$; the full-family improvement is established analytically. This branch does not exhaust higher-dimensional representations [5]. Exhaustiveness is unnecessary for an upper-bound competitor; we do not use it to claim a four-state or unrestricted minimum. The final certificate contains rational generators, stationary vectors, polynomial identities and rational enclosures of logarithms. Supporting checks cover the complete two-phase parametrisation, random binary partitions without renewal independence, the information inequality and the sharp limiting families. Numerical optimisations are recorded as discovery calculations, including failed runs, and are excluded from the logical certificate.

The unresolved problem is precise. We have proved a strict drop from the smallest exact architecture and a positive lower bound valid over all finite architectures. We have not determined $V_4(P_{1/20})$, $V_\infty(P_{1/20})$ or whether a finite model attains the latter. The fixed-statistics nonattainment theorem does not answer the fixed-law question. A further global certificate or an exactly compatible improving sequence is needed. These limits leave a complete stand-alone result: one extra hidden state can reduce the least physical cost of an unchanged stochastic signal, and the initial dwell density admits a sharp universal cost formula with a fully resolved attainment distinction.

Code and verification materials. The accompanying archive contains the manuscript source, exact certificates, reproduction scripts, figure code, numerical discovery record and bounded prior-art audit. The central rational certificate uses the Python standard library. Additional diagnostics and figures use NumPy, SciPy and Matplotlib. Internal independent checking is documented separately; it is not external peer review or proof-assistant verification.

Availability. This version (1.0.0-candidate) is archived at Zenodo, DOI 10.5281/zenodo.23721621, with source repository `https://github.com/ipitchford/hidden-architecture-exact-signal` and release page `https://evidencepress.org/releases/hidden-architecture-exact-signal/`.
Text and figures are released under CC0-1.0 and code under the MIT licence.

A Reset replacement and the scope of the three-state theorem

Lemma 7 (Reset replacement)**.** *Let a finite stationary chain realise an alternating-renewal law whose A dwell is* Exp(s)*. Replacing its whole A block by one reset state, as below, gives a chain with the same complete observed law and no larger entropy production. Consequently every exact realisation of P_q with at most two B states, and any number of A states, has cost at least $(1-q)\log(1/q)$.*

A single B state would give an exponential B dwell, which P_q excludes, so the lemma covers all realisations with exactly two B states. The reduction is a full-law corollary of the stationary-flux merge of Skinner and Dunkel [3, Supplementary Lemma 2]; it is included with attribution.

Proof. Keep $S = Q_{BB}$ and set

$$t_i = \sum_{a\in A} Q_{ia}, \quad r = \sum_{a,i} \pi_a Q_{ai} = \sum_i \pi_i t_i, \quad \alpha_i = r^{-1} \sum_a \pi_a Q_{ai}. \tag{65}$$

Replace the whole A block by one state with exit rate s and entrance distribution α into B. The resulting generator is (7). It is irreducible and has stationary distribution $(\pi(A), \pi_B)$ because $\pi(A) = r/s$ and $r\alpha + \pi_B S = 0$. Its B dwell density is the original $\alpha e^{\theta S} t$. The single reset state makes complete visits independent. Since independence is part of the target law, matching the two dwell laws and stationary fractions matches the full stationary process.

B-internal fluxes are unchanged. A-internal entropy is discarded, and each new boundary flux pair is a sum of old pairs. The log-sum inequality gives

$$J\left(\sum_a x_a, \sum_a y_a\right) \le \sum_a J(x_a, y_a), \tag{66}$$

so entropy cannot increase. The original chain need not be lumpable; this is an observationally equivalent replacement. Without renewal independence, resetting can destroy dependencies and does not generally preserve the observed law. Applying the reduction to P_q with at most two B states then invoking the proved three-state minimum gives the claimed extension. □

B Explicit finite parameters for the full-family improvement

The construction in Section 3.3 can be made fully explicit. Using $\nu = D/s$ and $z_0 = s/2$, define

$$\eta = \frac{\nu^4}{2(1+\nu)},$$
$$C_* = \left(q + z_0 + \nu + \frac{D}{2}\right)\left(\frac{\nu}{q} + \frac{D}{2z_0}\right)^2, \qquad M = 2 + \frac{4C_*}{\eta}. \tag{67}$$

All are positive rational functions of q. The first term of (42) gives $d_\infty \le -\eta$. For the symmetric-divergence Hessian,

$$v^{\mathsf{T}} \nabla^2 J(x,y) v = (x+y)\left(\frac{v_1}{x} - \frac{v_2}{y}\right)^2. \tag{68}$$

Apply the integral Taylor remainder to the last term of (40), from the baseline $(qM/2, z_0 M/2)$ with offset $(\nu, D/2)$. Along this segment the coordinates are at least their baseline values, and

their sum is at most $(M/2)(q + z_0 + \nu + D/2)$ because $M/2 \geq 1$. Equation (68) bounds the second directional derivative by $2C_*/M$. Therefore

$$0 \leq d_M - d_\infty \leq C_*/M < \eta/4, \qquad d_M < -3\eta/4. \tag{69}$$

Now put

$$B_* = 4\left[\left(q + \frac{s}{2}\right)^2 + \left(1 + \frac{s}{2}\right)^2 + \left\{\frac{s}{2}\left(M + 2 + \frac{\nu}{q}\right)\right\}^2\right],$$
$$\varepsilon_0 = \frac{2q}{s[q(M+2) + 2\nu]}, \qquad \varepsilon = \min\left\{\varepsilon_0, \frac{\eta}{2B_*}\right\} > 0. \tag{70}$$

For $0 \leq v \leq \varepsilon_0$, each of the six old directed fluxes remains at least half its value at zero. All decrease with v, and the sum of each old pair is initially one. The sums of their absolute slopes relative to their respective initial values are

$$q + s/2, \qquad 1 + s/2, \qquad (s/2)(M + 2 + \nu/q). \tag{71}$$

Equation (68) consequently gives $\Psi_M''(v) \leq B_*$. The new-edge terms are exactly linear. Taylor's theorem yields

$$\Psi_M(\varepsilon) - \Psi_M(0) \leq d_M\varepsilon + \frac{B_*}{2}\varepsilon^2 < -\frac{\eta\varepsilon}{2} < 0. \tag{72}$$

All rates are finite and strictly positive, and

$$\sigma(Q) < (1 - q)\log(1/q) - \frac{s\eta\varepsilon}{6}. \tag{73}$$

For rational q, every chosen parameter and generator entry is rational. These choices are deliberately conservative and only prove existence. The certified saving they give is extremely small: ε is of order 10^{-20} at $q = 10^{-4}$ and about 6×10^{-32} at $q = 0.999$. The quantitative size of the four-state saving is better judged from the explicit example (12) and from the numerical exploration in Section 6.

C Exact logarithm and polynomial certificates

For rational $x > 0$, put $v = (x - 1)/(x + 1)$. When $x > 1$,

$$\log x = 2\sum_{j=0}^{N-1} \frac{v^{2j+1}}{2j+1} + R_N, \qquad 0 < R_N \leq \frac{2v^{2N+1}}{(2N+1)(1-v^2)}. \tag{74}$$

All quantities on the right are rational. Reciprocal arguments or dyadic range reduction handle other positive rationals. The shipped certificate reduces each argument dyadically and then uses $N = 72$ series terms, giving rational enclosures of width below 10^{-24}; the enclosing rational endpoints are printed in the certificate output. Applying these bounds to

the five terms in (32) and to $(19/20)\log 20$ certifies (33) without a floating-point comparison. Generator row sums, stationary balance, the annihilator (30) and the transfer-polynomial cancellation are checked with rational arithmetic.

The two-phase numerical diagnostic independently calculates costs from the full stationary fluxes, the cycle-current expression and (23), including parameters near equilibrium. These checks guard the algebraic transcription; the series inequality in the text, rather than a parameter grid, proves global optimality.

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